Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/06/2005
PI Name:	Orwoll, Robert Ph.D.		
Project Title:	Polyarylene Ether and Polymide Materials with Improved Radiation Shielding Capabilities		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Williamsburg	**State:**	VA
Zip Code:	23187-8795	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	12/01/2002	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1900		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To develop novel, high-strength plastics with chemical compositions which have been modified from those in use today so that they have the added benefit of reducing the effects of cosmic radiation in the interior of spacecraft and lunar and Martian habitats. Collisions between the high-energy (HZE) particles in cosmic radiation with atoms in the metal shell of a spacecraft result in nuclear fission which can lead to dangerous doses of radiation inside of a spacecraft. However, materials having high numbers of hydrogen atoms per unit mass are known to be effective shields. In this task, high-strength plastic materials are being formulated and prepared with large quantities of hydrogen. For this work , aromatic polyimides and polyarylene ethers were selected because of their thermal stability and mechanical strength. To provide light-weight polyimide and polyarylene ether materials for (i) shielding from cosmic radiation inside spacecraft and extraterrestrial habitats while (ii) also serving other functions possibly including structural support, thermal and acoustic insulation, micrometeorite shielding, plumbing, and space suits. Plastic components in the living and working quarters of spacecraft will add to the protection of both humans and electronic devices while simultaneously serving as		

	part of the spacecraft's structure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Potential materials that might provide radiation shielding for high-flying aircraft.
Task Progress:	Nine polymers have been synthesized, six aromatic polyimides and three polyarylene ethers—all with high molecular weights. These are hybrid materials with molecular structures that contain both (i) hydrogen-rich methyl groups for enhanced radiation shielding and (ii) aromatic units in the backbone for favorable mechanical and thermal properties. These materials contain no elements higher than oxygen—an important consideration given that nuclear fragmentation from collisions between HZE particles and nuclei in the shield worsens with higher atomic-numbered elements. The hydrogen contents of the nine materials range from 0.041 to 0.069 moles of hydrogen per gram—intermediate between 0.026 for the high-performance, space-tested Kapton and 0.143 for non-aromatic polyethylene. Four of these polymers were submitted to the Brookhaven National Laboratory this year for testing in the accelerator for their effectiveness as shielding materials. Sufficient quantities of the remaining five polymers have now been synthesized for targets for future runs at Brookhaven. In addition targets were prepared from three commercial polymers that have aromatic units and high hydrogen content and submitted for radiation testing. Glass transition temperatures, thermal degradation, inherent viscosities, and solubility profiles have been measured for the nine polymers synthesized for this program. Measurements of additional thermal and mechanical properties are underway. While we have some results from runs at Brookhaven this summer, we await results from the September runs. When we have these, we will collaborate with modelers at NASA-LaRC to interpret the Brookhaven measurements on our formulations and to make comparisons with other candidate shielding materials.
Bibliography Type:	Description: (Last Updated: 03/30/2017)
Articles in Other Journals or Periodicals	Hu LY, Yang S, Park CS, Orwoll, RA, Jensen BJ. "Synthesis of polyimide materials for radiation shielding." Polymer Preprints 2004;45(2):673-674. , Aug-2004
Presentation	Hu, L. Y.; Miller, A. E.; Park, C. S.; Plichta, K. A.; Rochford, S. J.; Schulz, M. E.; Yang, S.; Orwoll, R. A. "Aliphatic/Aromatic Hybrid Polymes for Functionally Graded Radiation Shielding" Space 2004 Conference and Exposition (American Institute of Aeronautics and Astronautics), San Diego, Paper AIAA-2004-6030 Sep-2004